PROJECT 17 · FINAL SUBMISSION

Next-Day Wildfire Spread Prediction

Tabular machine learning and spatial deep learning, compared with care for leakage

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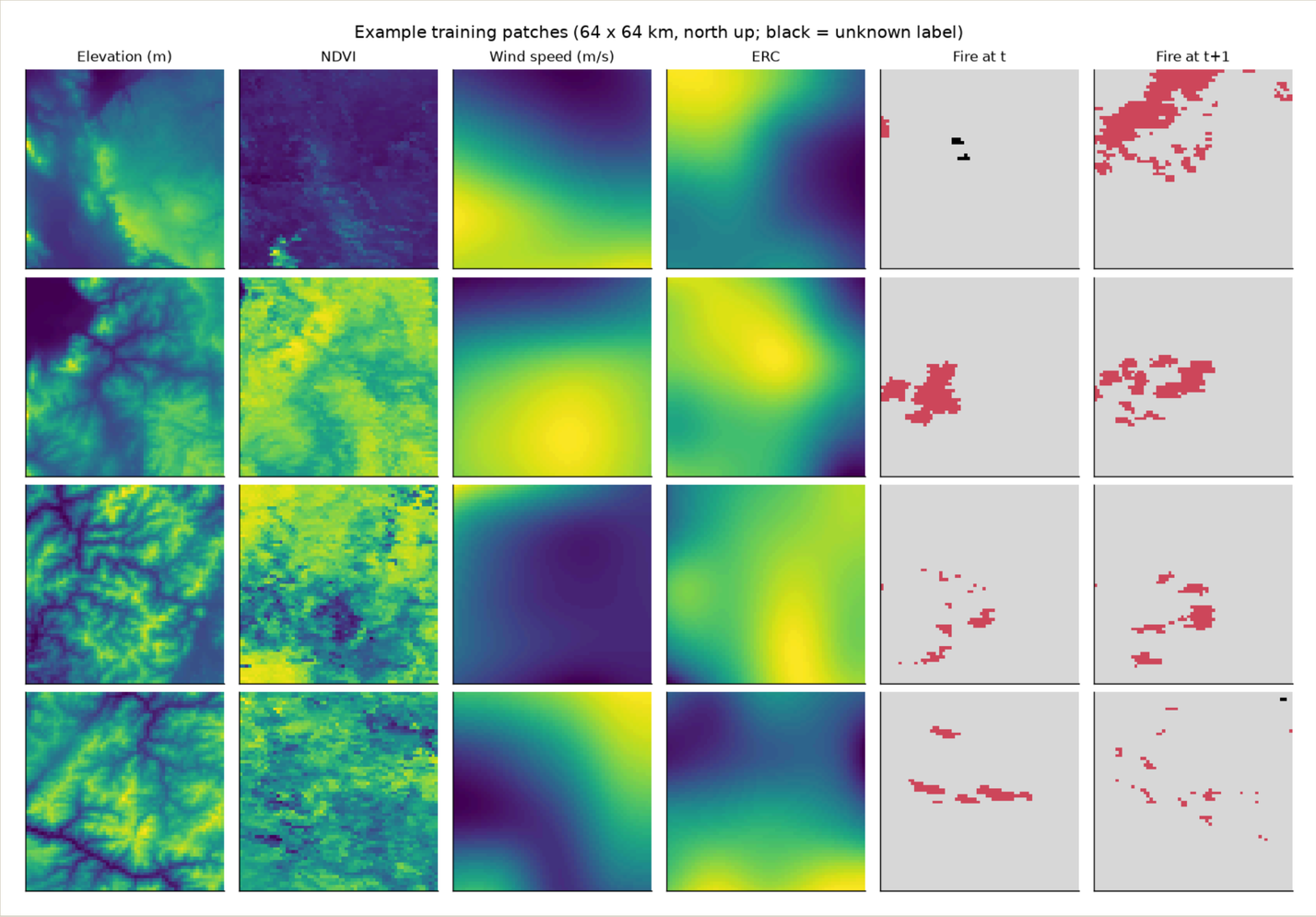
THE TASK

Which pixels will be burning tomorrow?

- For each 1 km pixel around an active fire, estimate the chance it burns on day t+1
- Inputs: 11 environmental rasters and the fire map from day t
- Only 1.10% of training pixels are positive, so accuracy is useless. We use PR-AUC
- Two families compared on the same pixels: trees on hand-made features, and CNN and U-Net on images

DATA

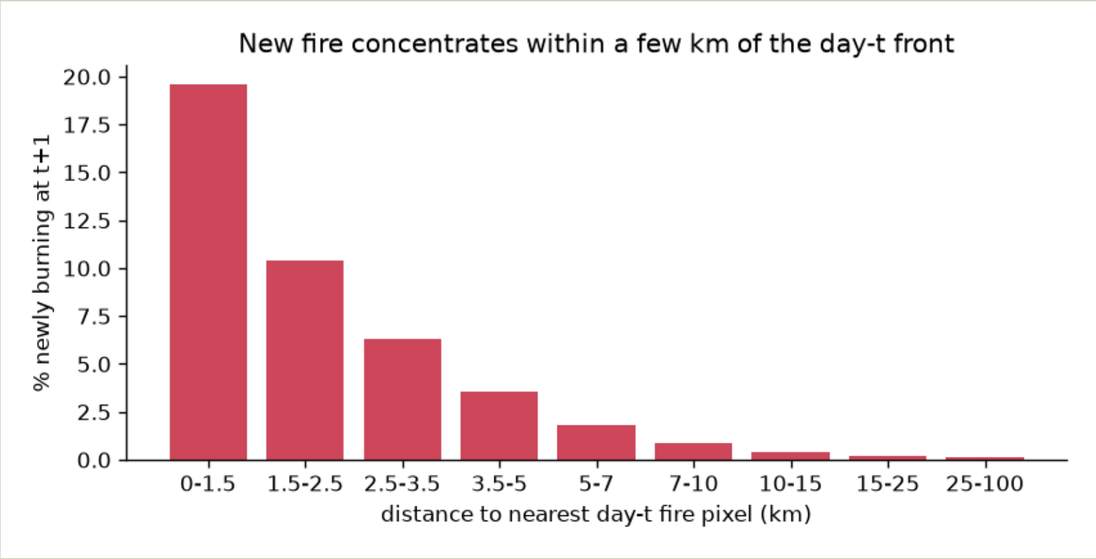
NDWS: one day of inputs, the next day's fire

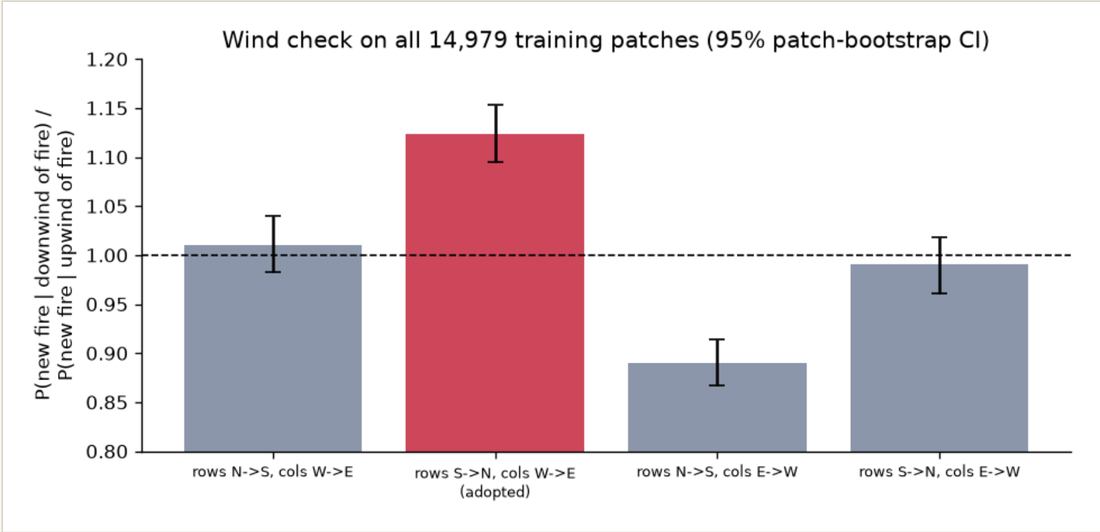


- 18,545 patches of 64 x 64 km, US, 2012 to 2020
- Labels: fire, no fire, or unknown (excluded)
- No coordinates, dates or event IDs in the release

WHAT THE DATA SHOWS

Fire mostly appears close to fire





74% of next-day fire pixels were not burning on day *t*. Persistence alone recalls 26% of them. New fire is 1.12 times as likely downwind as upwind (95% CI 1.09 to 1.15).

HONEST EVALUATION

Three test splits, three different questions

Protocol	Weeks kept apart	Terrain kept apart	Train / val / test patches
Official	Yes	No	14,979 / 1,877 / 1,689
Spatial	No	Yes	14,822 / 1,855 / 1,868
Strict	Yes	Yes	12,050 / 389 / 405

Identical terrain shows about **95%** of official test patches overlap training locations. The strict test set is small (405 patches), so its intervals are wide.

A HIDDEN SHORTCUT

Patches are centred on fire, so position leaks the answer

- On full patches, a prior using only position scores 0.137 on strict, above every valid model
- A U-Net trained on full patches gains +0.084, +0.103 and +0.089 over random crops
- So networks train on random 32 x 32 crops
- Every model is also scored on fixed random windows where the centre is not special

LightGBM test PR-AUC	Official	Spatial	Strict
Features	0.263	0.138	0.124
+ position	0.345	0.278	0.251

Adding position helps a lot, so the tabular features do not already contain all of it.

MODELS

Trees, networks and hybrids

Family	What it sees	Models
Baselines	Trivial rules	Dummy, persistence, neighbourhood persistence
Tabular	32 features per pixel	Logistic regression, random forest, XGBoost, LightGBM
Spatial	Image patches	Compact CNN (81,097 params), U-Net (1,086,841 params)
Hybrid	Both	U-Net with 31 channels, and late fusion of tree and network

U-Net settings were chosen on a separate tuning split: weighted cross-entropy (weight 10), dropout 0.2. Fusion weights use half of validation, and the other half picks the hybrid.

RESULTS

No network or hybrid beats the best tree

Test PR-AUC [95% CI]	Official	Spatial	Strict
Dummy	0.013 [0.012, 0.014]	0.005 [0.005, 0.005]	0.006 [0.005, 0.007]
Random forest	**0.265 [0.248, 0.282]**	0.135 [0.123, 0.149]	0.120 [0.097, 0.145]
XGBoost	**0.265 [0.248, 0.283]**	**0.141 [0.128, 0.154]**	0.115 [0.094, 0.141]
LightGBM	0.263 [0.246, 0.281]	0.138 [0.127, 0.150]	**0.124 [0.103, 0.148]**
Compact CNN	0.256 [0.240, 0.272]	0.129 [0.117, 0.141]	0.093 [0.071, 0.119]
U-Net	0.257 [0.242, 0.272]	0.130 [0.118, 0.142]	0.101 [0.079, 0.124]
U-Net + 31 channels	0.254 [0.239, 0.270]	0.129 [0.117, 0.141]	0.116 [0.091, 0.141]
Hybrid (selected on validation)	0.259 [0.244, 0.276]	0.140 [0.127, 0.152]	0.115 [0.093, 0.140]

Positive rates differ across splits (1.25%, 0.48%, 0.65%), so compare within a column. Network rows use the first of three training seeds.

ARE THE GAPS REAL?

Paired tests against the validation-selected tree (XGBoost)

Difference in PR-AUC	Official	Spatial	Strict
U-Net - XGBoost	-0.008	-0.011	-0.014
CNN - XGBoost	-0.009	-0.012	-0.022
Hybrid - XGBoost	-0.006	-0.001	-0.001
Hybrid - U-Net	+0.003	+0.010	+0.014

- U-Net and CNN are significantly below XGBoost on all three splits
- The hybrid is significantly below XGBoost on official, and indistinguishable on spatial and strict
- The hybrid significantly improves on the U-Net on spatial and strict

WITHOUT THE CENTRE

Fixed windows close the gap on official and spatial, not on strict

Test PR-AUC on fixed windows	Official	Spatial	Strict
LightGBM (window-trained)	0.290	0.169	0.148
U-Net	0.297	0.167	0.134
Hybrid	0.301	0.172	0.139
Position prior only	0.032	0.015	0.020

- Official: the hybrid beats the window-trained tree by +0.012, the only significant hybrid win over a tree
- Strict: the U-Net trails by -0.015 and the hybrid by -0.009

STABILITY

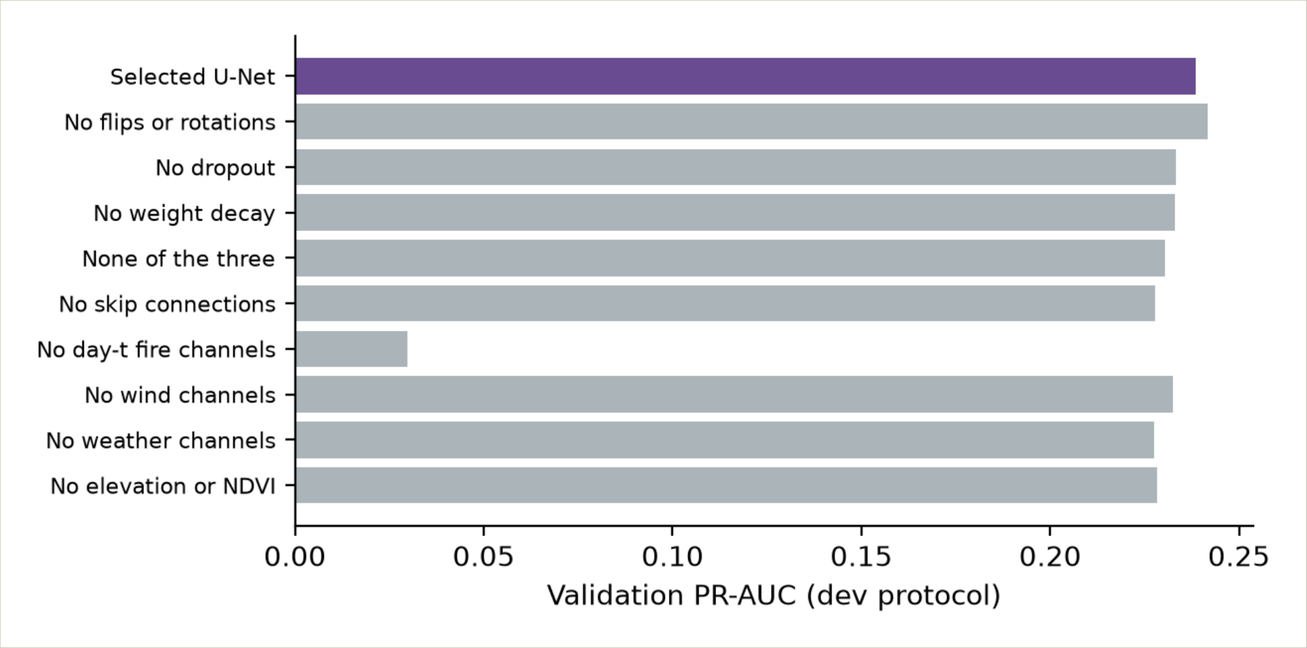
Three seeds: small gaps are within training noise

Mean test PR-AUC	Official	Spatial	Strict
Compact CNN	0.255	0.127	0.099
U-Net	0.252	0.128	0.104
U-Net + 31 channels	0.254	0.135	0.116

Seed standard deviation runs from 0.001 to 0.013. The 31-channel U-Net's lead on strict has the largest spread, so we treat it as suggestive only.

WHAT MATTERS

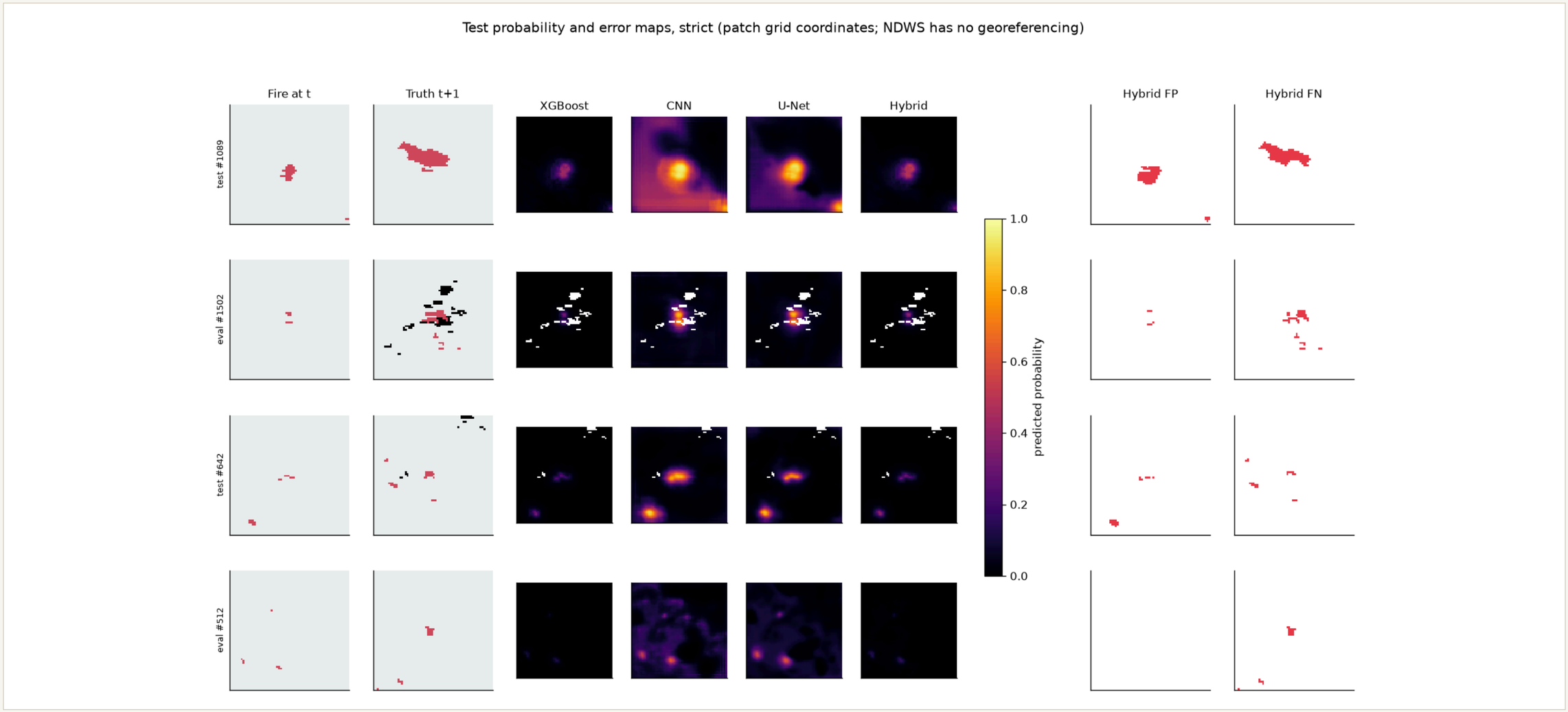
Today's fire map dominates



- Without day-*t* fire channels: validation PR-AUC 0.239 falls to 0.030
- Tabular on strict: raw values only 0.012, fire context only 0.095
- Removing augmentation gave 0.242, so we do not claim it helped

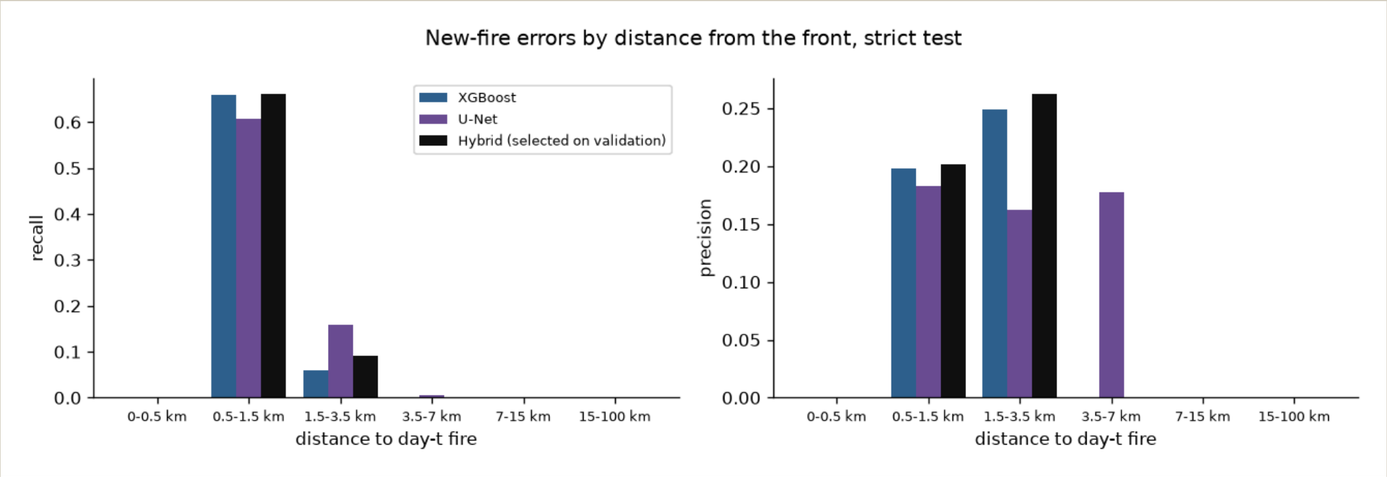
PREDICTION MAPS

Actual, trees, networks, hybrid, and errors (strict test)



WHERE MODELS FAIL

Fire far from today's perimeter is missed



XGBoost recall	Under 1.5 km	1.5-3.5	3.5-7
Official	0.95	0.45	0.01
Spatial	0.72	0.19	0.01
Strict	0.66	0.06	0.00

SINCE PHASE 1

What changed, and why

- Graded Phase 1 materials are frozen in `phase1_submission/`, and the official and spatial results reproduce to three decimals
- New **strict split**: the Phase 1 spatial split still mixed weeks between training and test
- New **position-leak test**: Phase 1 did not test whether tabular models exploit position
- New **Dummy baseline, CNN, tuned U-Net and hybrids** to complete the comparison

LIMITS

What this work cannot claim

- Weeks are randomly assigned, so this is not forecasting into future seasons
- No coordinates, dates, fuel moisture or suppression data
- Terrain matching only finds exact overlaps, so nearby fires may still cross splits
- The strict test has only 405 patches
- Samples exist only where fire is present, so this is not monitoring of all land

CONCLUSION

Well-built features still lead on this benchmark

- Best test PR-AUC: XGBoost 0.265 (official) and 0.141 (spatial), LightGBM 0.124 (strict)
- The hybrid improved on the U-Net but did not beat the best tree
- Benchmark scores depend strongly on how data is split and sampled
- Next: WildfireSpreadTS, for real dates and multi-day models